Experiment 5-Fourier Series Analysis of Periodic Signals

OBJECTIVE :

i). To analyze Fourier series of continuous-time periodic signals and analyze their waveform characteristics.

ii).To reconstruct periodic signals using Fourier series and verify the reconstructed waveform with the original signal.

i) To analyze Fourier series of continuous-time periodic signals and analyze their waveform Characteristics.

Software / Laboratory Environment:

MATLAB

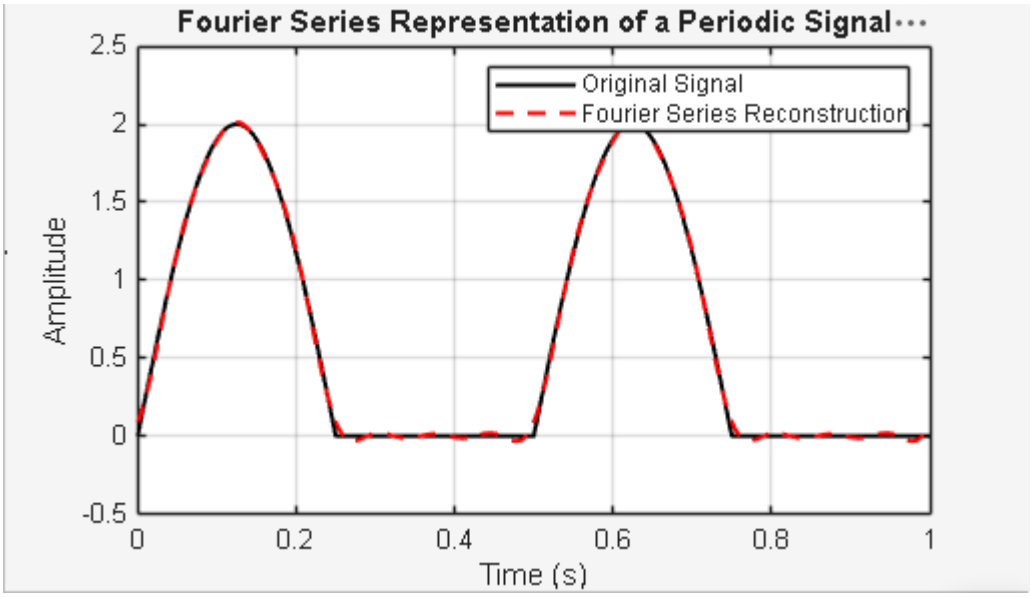
MATLAB Coding:

```
clc;
clear;
close all;
A = 2;
f0 = 2;
T = 1/f0
t = linspace(0, 2*T, 2000)
x = A*sin(2*pi*f0*t);
x(x < 0) = 0;
N = 7;
tc = linspace(0, T, 1000);
xc = A*sin(2*pi*f0*tc);
xc(xc < 0) = 0;
a0 = (2/T)*trapz(tc, xc)/2;
an = zeros(1, N);
bn = zeros(1, N);
for k = 1:N
    an(k) = (2/T)*trapz(tc, xc.*cos(2*pi*k*f0*tc));
    bn(k) = (2/T)*trapz(tc, xc.*sin(2*pi*k*f0*tc));
end
xFS = a0*ones(size(t));

for k = 1:N
    xFS = xFS + an(k)*cos(2*pi*k*f0*t) ...
        + bn(k)*sin(2*pi*k*f0*t);
end
figure;
```

```
plot(t, x, 'k', 'LineWidth', 1.5);
hold on;
plot(t, xFS, 'r--', 'LineWidth', 1.5);
grid on;
xlabel('Time (s)');
ylabel('Amplitude');
title('Fourier Series Representation of a Periodic Signal');
legend('Original Signal', 'Fourier Series Reconstruction');
```

Output Graphs:



Conclusion:

The Fourier series representation of periodic continuous-time signals was successfully analysed and the Fourier coefficients and harmonic components were determined. The amplitude and phase spectra were generated and their characteristics were studied. The periodic signal was reconstructed using Fourier series and the reconstructed waveform was verified with the original signal.

Rubrics:

Objective	Assessment Criterio	Excellent – 4	Good – 3	Satisfactory – 2	Needs Improvemen	Weightage	Marks(10)	Marks obt

	n				t – 1			ained
(i) To analyze the Fourier series of continuous-time periodic signals and analyze their waveform characteristics.	Fourier Series Analysis	Correctly determines the fundamental period and frequency and computes the Fourier coefficients for every given periodic signal, analytically or in MATLAB. Uses suitable commands with	Computes the coefficients for most signals correctly, with minor errors in limits, algebra, code or parameters.	Computes the coefficients for only some signals, or has noticeable errors in integration, limits or code, and needs guidance.	Unable to compute the Fourier coefficients of the given signals.	20	2	

		correct limits and obtains accurate coefficient values.						
	Waveform Characteristics & Spectrum Analysis	Correctly identifies symmetry (even, odd, half-wave), DC value and the harmonics present. Plots the amplitude and phase spectra with titles, axis labels	Identifies most waveform characteristics and plots the spectra correctly, with minor errors or limited explanation.	Identifies basic characteristics. Spectra have noticeable errors or the explanation is incomplete, and needs guidance.	Unable to identify the waveform characteristics or plot the spectra.	20	2	

		and grid. Explains how symmetry and the shape of the waveform determine the harmonic content and the rate of decay of the coefficients.						
(ii) To reconstruct periodic signals using the Fourier series and verify the reconst	Signal Reconstruction using Fourier Series	Correctly synthesizes each periodic signal as a partial sum of N harmonics in	Reconstructs the signals correctly with minor errors in code, time range, parameters	Reconstructs only some signals or uses a single value of N, with noticeable errors	Unable to reconstruct the signals using the Fourier series.	20	2	

ructed wavefo rm with the origina l signal.		MATL AB, using a suitabl e time vector and loop or vectori zed code. Recons tructs with several values of N and plots them clearly against the origina l.	or labelli ng, and uses fewer values of N.	in code or plots, and needs assista nce.				
	Verifica tion of Recons tructed Wavefo rm	Verifie s the reconst ruction against the origina l wavefo rm using	Verifie s the reconst ruction with overla y plots and some error measur	Perfor ms basic visual compar ison only, with noticea ble errors	Unable to verify the reconst ructed wavef orm against the origina	20	2	

		overlay plots and numerical error for each N. Shows that the error decreases as N increases, and identifies the Gibbs phenomenon at discontinuities with a correct explanation.	e, with minor errors in calculation or interpretation of convergence.	or no numerical error. Convergence or Gibbs phenomenon is not explained, and needs guidance.	l signal.			
Interpret the Fourier series results and maintai	Engineering Analysis, Record &Interpretation	Provides technically accurate interpr	Maintains a proper record with mostly accurat	Record and interpretation are basic or	Incomplete record, limited or incorrect	20	2	

n the record.		etation of the results, relating the number of harmonics, symmetry and discontinuities to reconstruction accuracy. Maintains a complete, well-organized record with code, plots, observations and conclusions.	e interpretation, with minor omissions in observations or conclusions.	partly incomplete, with limited technical reasoning.	interpretation of the results.			
					Total	100	10	